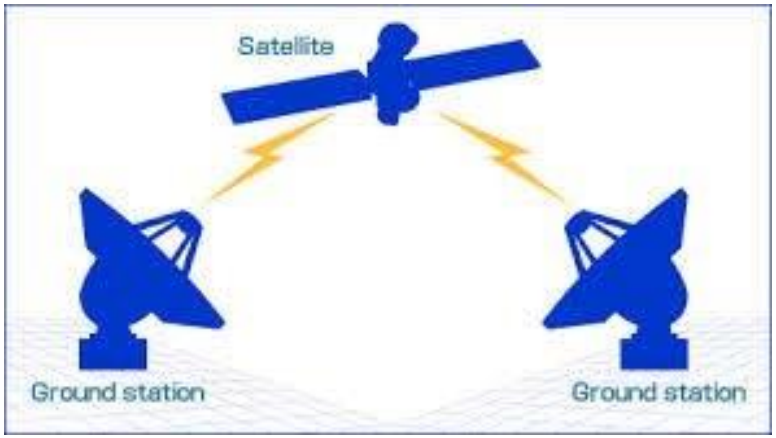
Telecommunication Systems
ICE 3109

Lecture 13
Satellite Communications

Introduction

- How do Satellites Work
  - Two Stations on Earth want to communicate through radio broadcast but are too far away to use conventional means.



  - The two stations can use a satellite as a relay station for their communication
  - One **Earth Station** sends a transmission to the satellite.
    - This is called a **Uplink**.
  - The satellite **Transponder** converts the signal and sends it down to the second earth station.
    - This is called a **Downlink**.

Introduction

- Satellite is in (earth) orbit
  - Special orbits have particularly useful properties
  - Carries its own source of power
- Communications possible with:
  - Ground station fixed on earth surface
  - Moving platform
  - Another orbiting satellite

Introduction

- Advantages of Satellites
  - High channel capacity (>100 Mb/s)
  - Low error rates ($P_e \sim 10^{-6}$)
  - Stable cost environment (no long-distance cables or national boundaries)
  - Wide area coverage (whole North America, for instance)
  - Coverage can be shaped by antenna patterns

Introduction

- Disadvantages of Satellites
  - Expensive to launch
  - Expensive ground stations required
  - Cannot be maintained
  - Limited frequency spectrum
  - Limited orbital space (geosynchronous)
  - Constant ground monitoring required for positioning and operational control

Introduction

- Satellite Characteristics
  - Space vehicle used as communications platform
    - Earth-Space-Earth, Space-Earth, Space-Space
  - Space vehicle used as sensor platform with communications
  - Ground station(s) (Tx/Rx)
  - Ground receivers (Rx only)
  - Orbital parameters
    - Height, Orientation, Location, etc
  - VHF, UHF, and microwave radiation used for communications with Ground Station(s)
  - Power sources
    - Principally solar power

Introduction

- Applications
  - Telecommunications
  - Military communications
  - Navigation systems
  - Remote sensing and surveillance
  - Radio / Television Broadcasting
  - Astronomical research
  - Weather observation

Introduction

- Growth of Communication Satellite
  - 1957, Russia , Sputnik-1
  - 1958, NASA, Score
  - 1962, Teslar-1
  - 1963, Syncom-1
  - 1965, Intelsat-1

Introduction

- Frequency Band Available for Satellite communication

Band	Frequency Range	Total Bandwidth	General Application
L	1 to 2 GHz	1 GHz	Mobile satellite service (MSS)
S	2 to 4 GHz	2 GHz	MSS, NASA, deep space research
C	4 to 8 GHz	4 GHz	Fixed satellite service (FSS)
X	8 to 12.5 GHz	4.5 GHz	FSS military, terrestrial earth exploration, and meteorological satellites
Ku	12.5 to 18 GHz	5.5 GHz	FSS, broadcast satellite service (BSS)
K	18 to 26.5 GHz	8.5 GHz	BSS, FSS
Ka	26.5 to 40 GHz	13.5 GHz	FSS

Introduction

- Kepler's Laws for Orbits

Kepler's Laws (1609 for 1,2; 1619 for 3) were based on observations of the motions of planets.

(1) All planets travel in elliptical orbits with the Sun at one focus.
→ defines the shape of orbits

(2) The radius from the Sun to the planet sweeps out equal areas in equal times.

→ determines how orbital position varies in time

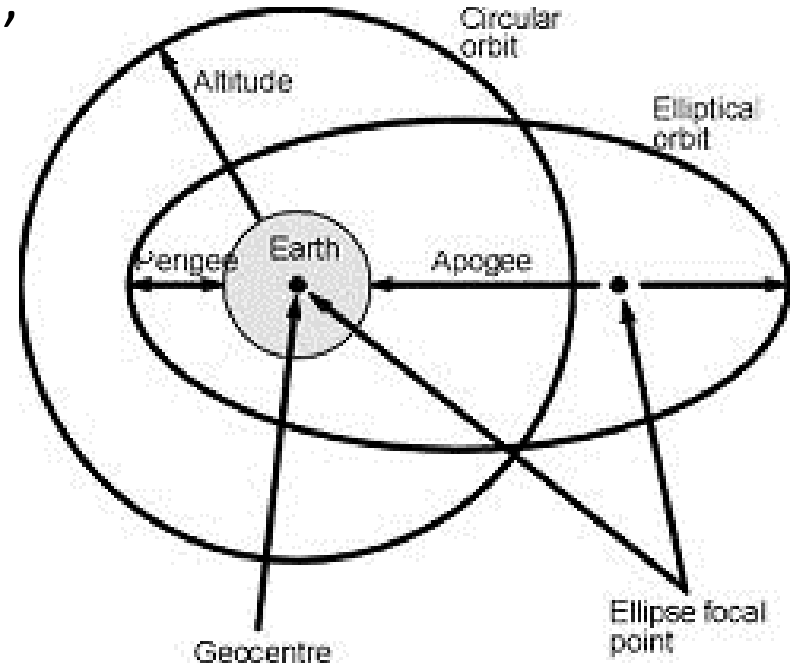
(1) The square of the period of a planet's revolution is proportional to the cube of its semimajor axis.

→ suggests that there is some systematic factor at work

Satellite Orbits

- Most of the satellites are called orbital satellites
  - Non synchronous
  - On the basis of rotation pattern,

    2 Types
  - Circular orbit
    - Fixed speed or rotation
    - the distance from the Earth remains the same at all times
  - Elliptical orbit
    - Speed depends on the height from the earth
    - Elliptical orbit changes the distance to the Earth



Satellite Orbits

Direction of rotation around the Earth:

There are two ways in which a satellite orbit may be categorized:

Posigrade:

The rotation around the earth is said to be posigrade when it rotates in the same direction as the rotation of the Earth.

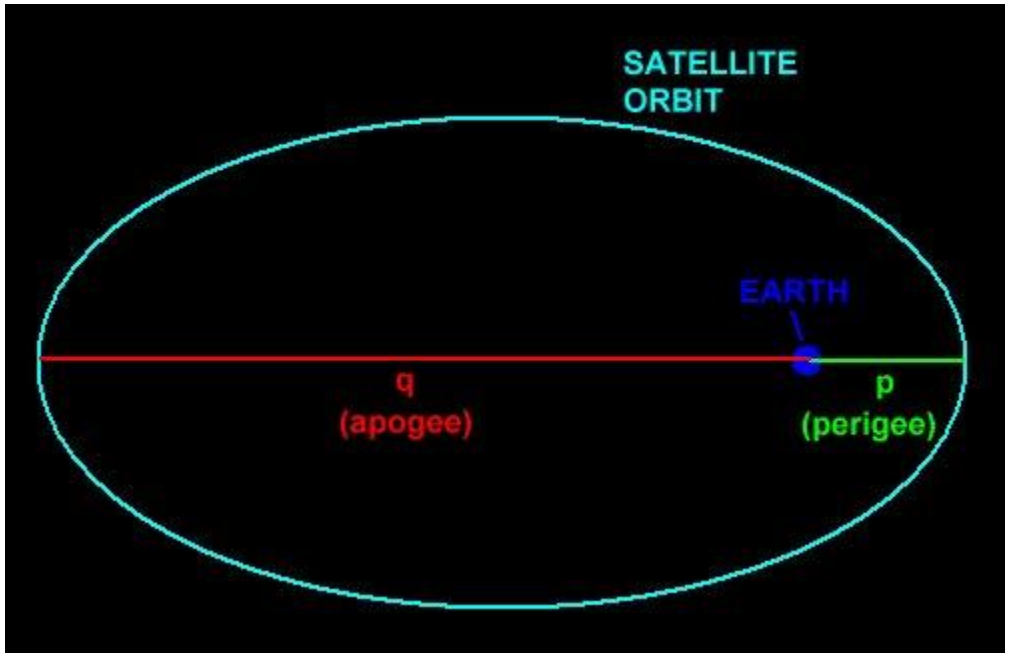
Retrograde:

The rotation around the earth is said to be retrograde when it rotates in the opposite direction to the rotation of the Earth

Orbit Parameters

Apogee (q) refers to the furthest distance that the satellite can have from the Earth. It can also refer to that point in the satellite's orbit at which the satellites reaches the furthest distance.

Perigee (p) refers to the closest distance that the satellite can have from the Earth. It can also refer to that point in the satellite's orbit at which the satellites reaches the closest distance.

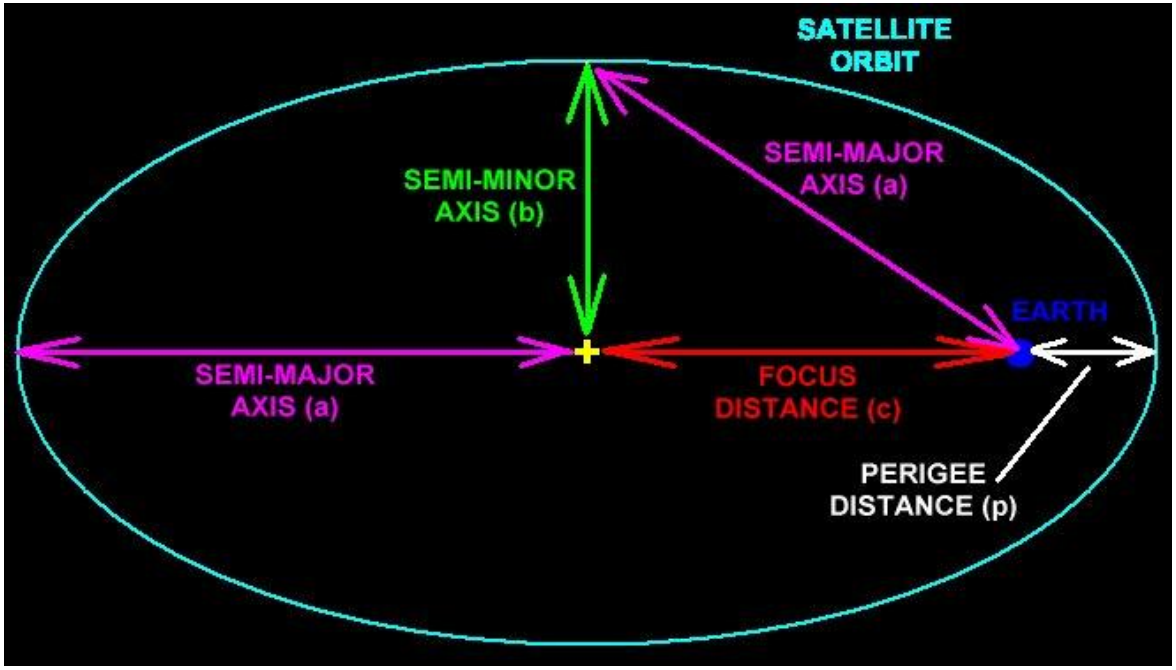


Orbit Parameters

Semi-Major Axis (a): The distance from the center of the orbit ellipse to satellite's apogee or perigee point. This is also defined as the average distance of the satellite from the Earth's center.

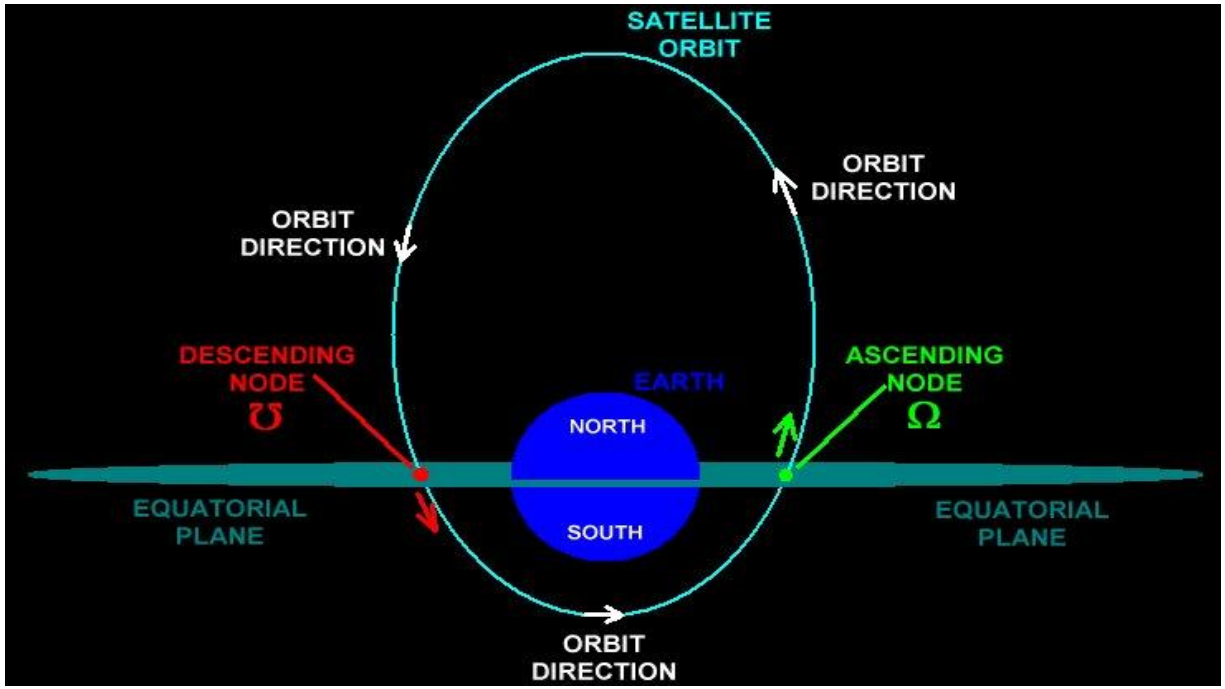
Semi-Minor Axis (b): The shortest distance from the true center of the orbit ellipse to the orbit path.

Focus Distance (c): The distance from the true center of the orbit ellipse to the center of the Earth.



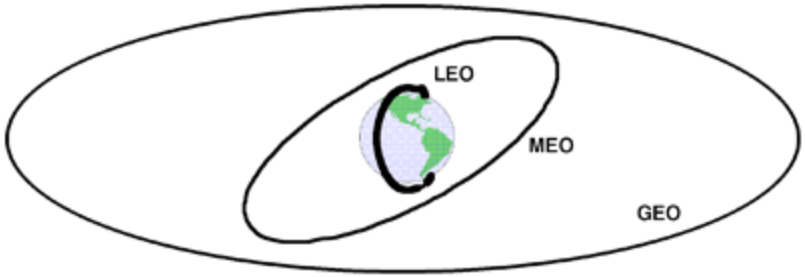
Orbit Parameters

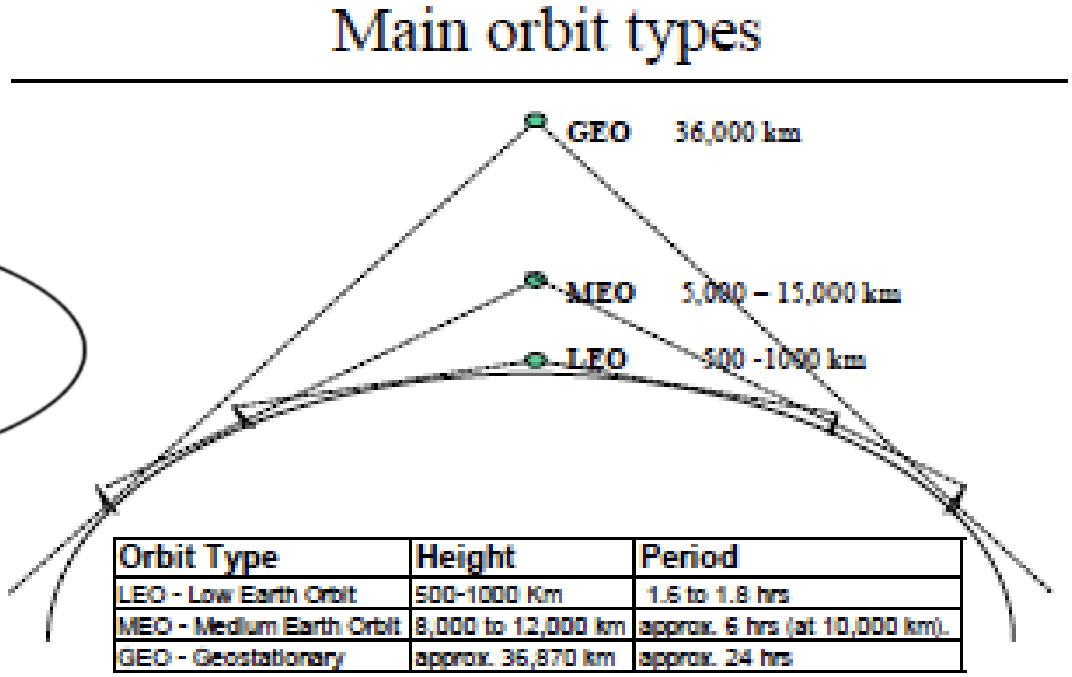
Period (T): The time required for the satellite to orbit the Earth once.
Ascending Node (Ω): The precise point in a satellite's orbit that intersects the equatorial plane of the Earth as the satellite moves from the southern to the northern hemisphere (ascending).
Descending Node (℧): The precise point in a satellite's orbit that intersects the equatorial plane of the Earth as the satellite moves from the northern to the southern hemisphere (descending).



Satellite Orbits

- Satellite Elevation Categories
  - LEO
  - MEO
  - GEO





Orbit Type	Height	Period
LEO - Low Earth Orbit	500-1000 Km	1.6 to 1.8 hrs
MEO - Medium Earth Orbit	8,000 to 12,000 km	approx. 6 hrs (at 10,000 km).
GEO - Geostationary	approx. 36,870 km	approx. 24 hrs

Satellite Orbits

- Low Earth Orbit (LEO)
  - 1Ghz to 2.5 GHz frequency
  - 80 - 500 km altitude
  - Lower path loss (lower power → Small antenna → less weight.
- Medium Earth Orbit (MEO)
  - 1.2 GHz to 1.66 GHz
  - 2000 - 35000 km altitude
  - Van Allen radiation between 200 - 1000 km
- Geostationary Orbit (GEO)
  - 2GHz to 18 GHz
  - 35,786 km altitude (42,164.57 km radius)
  - Difficult orbital insertion and maintenance

Satellite Orbits

- Low Earth Orbit Satellite (LEO)
  - LEO satellites are much closer to the earth than GEO satellites, ranging from 500 to 1,500 km above the surface.
  - LEO satellites don't stay in fixed position relative to the surface, and are only visible for 15 to 20 minutes each pass.
  - A network of LEO satellites is necessary for LEO satellites to be useful
  - A LEO satellite's proximity to earth compared to a GEO satellite gives it a better signal strength and less of a time delay, which makes it better for point to point communication.
  - A LEO satellite's smaller area of coverage is less of a waste of bandwidth.

Satellite Orbits

- Medium Earth Orbit (MEO) Satellite
  - MEO satellites are similar to LEO satellites in functionality.
  - MEO satellites are visible for much longer periods of time than LEO satellites, usually between 2 to 8 hours.
  - MEO satellites have a larger coverage area than LEO satellites.
  - A MEO satellite’s distance gives it a longer time delay and weaker signal than a LEO satellite, though not as bad as a GEO satellite.

Satellite Orbits

- Geostationary Earth Orbit (GEO)
  - These satellites are in orbit 35,863 km above the earth's surface along the equator.
  - Objects in Geostationary orbit revolve around the earth at the same speed as the earth rotates.
    - This means GEO satellites remain in the same position relative to the surface of earth.
  - Each satellite can cover 120 degrees latitude
  - Orbital Radius = 42,164.17 km
  - Earth Radius = 6,378.137 km (avg)
  - Period (Sidereal Day) = 23.9344696 hr (86164.090530833 seconds)
  - Velocity 6840 mph (11033 km/h)
  - Round trip time delay 238 ms

Satellite Orbits

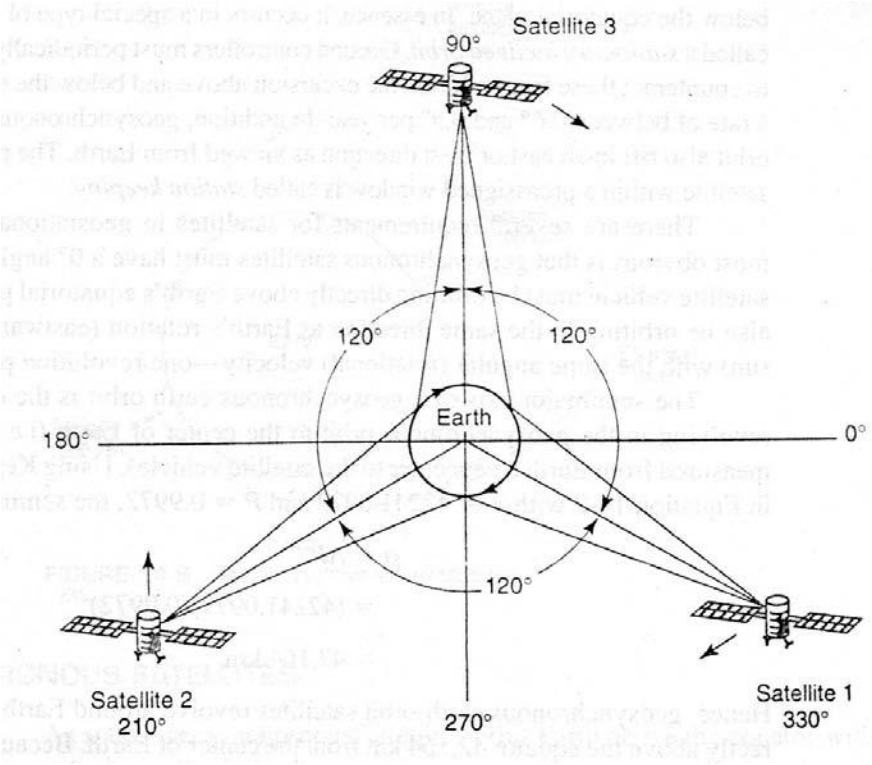
- Geostationary Earth Orbit (GEO)
  - A GEO satellite's distance from earth gives it a large coverage area, almost a fourth of the earth's surface.
  - EO satellites have a 24 hour view of a particular area.
    - These factors make it ideal for satellite broadcast and other multipoint applications.
  - A GEO satellite's distance also cause it to have both a comparatively weak signal and a time delay in the signal, which is bad for point to point communication.
  - GEO satellites, centered above the equator, have difficulty broadcasting signals to near polar regions

Satellite Orbits

- Clarke Orbit
  - Geosynchronous orbit also known as Clarke orbit
    - Famous Engineer, scientist, and a science fiction writer Arthur C. Clarke
    - Specification
      - Located directly above the equator
      - Travel in the same direction to the earth at a 6840 mph
      - Altitude 22300 miles
      - Complete one revolution in 24 hr.
    - An international agreement
      - Satellites in Clarke orbit must be separated by at least 1833 miles
      - Angular separation of 4 degree

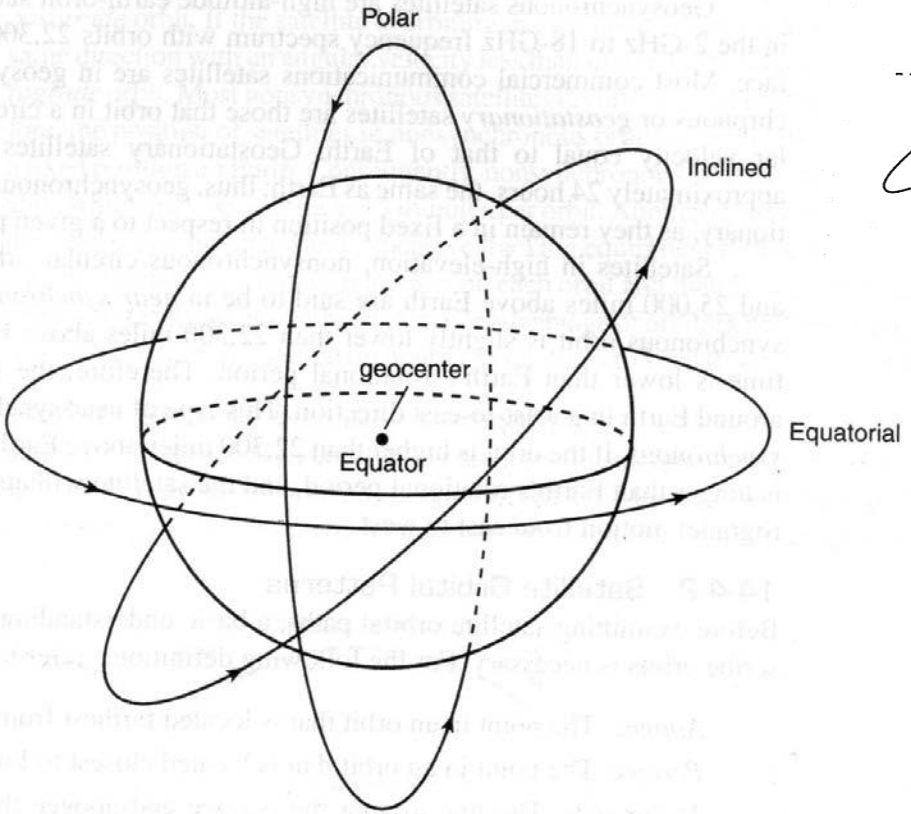
Satellite Orbits

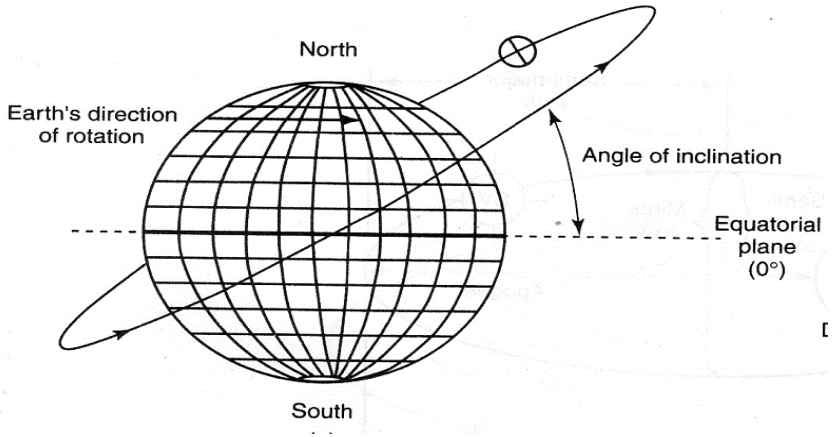
- Clarke orbit
  - Three satellites in Clarke orbit separated by 120 degree cover the whole globe shown in Fig.

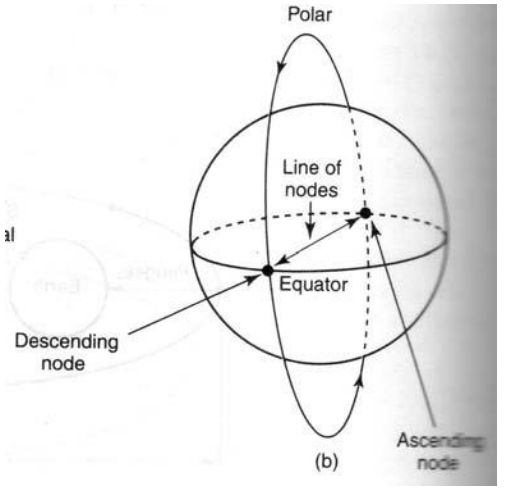


Satellite Orbits

- Orbital Pattern

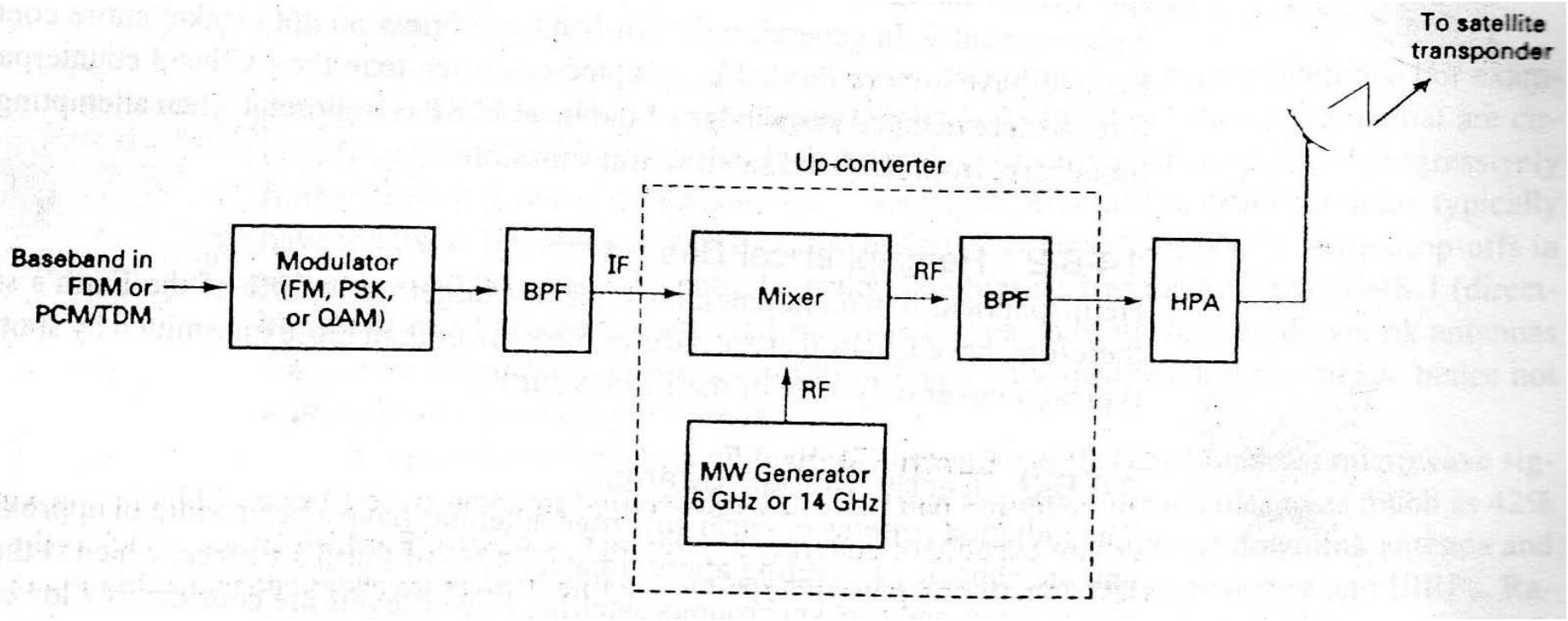






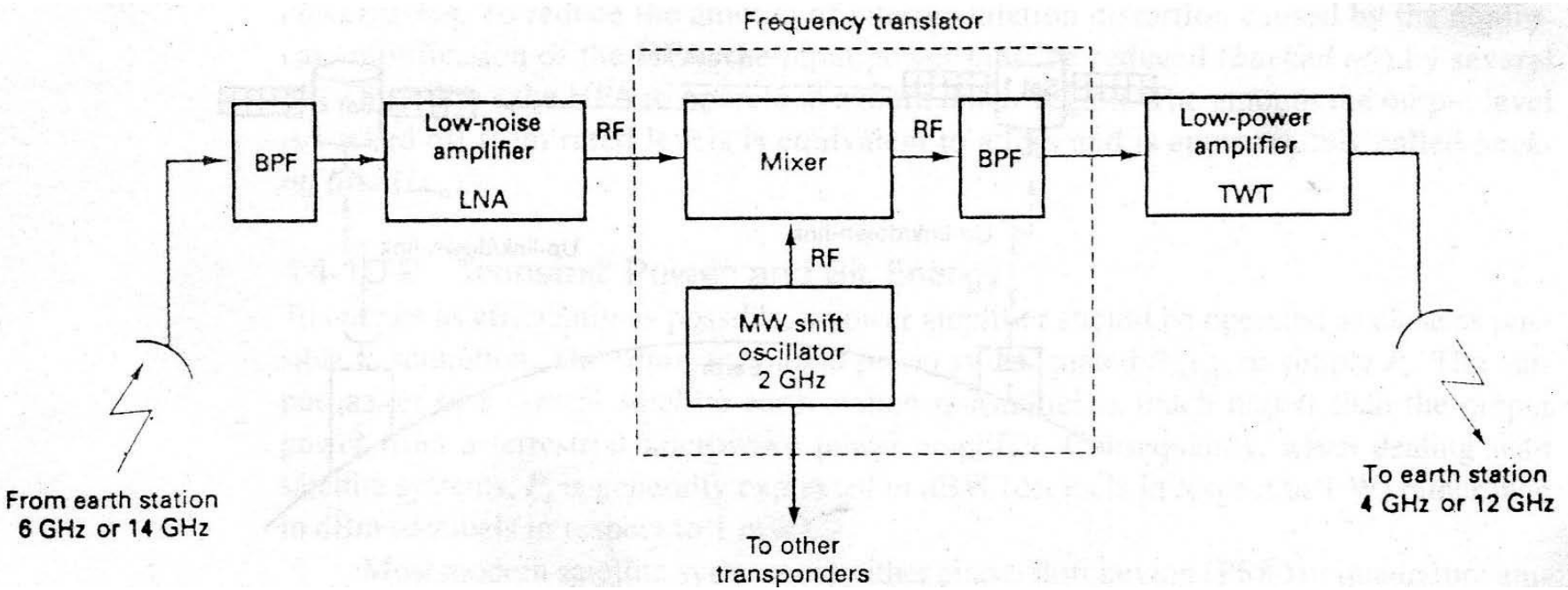
Satellite System Link Models

- Up link model



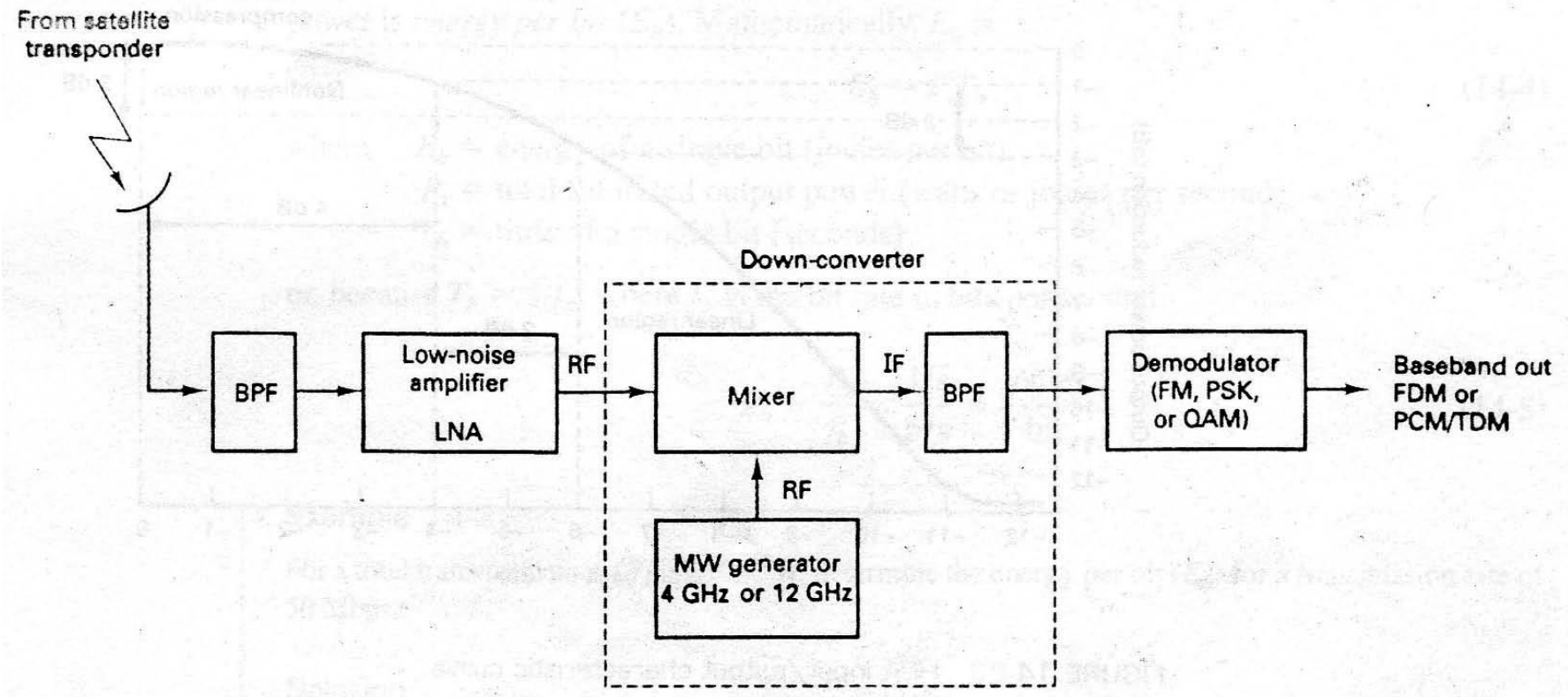
Satellite System Link Models

- Transponder



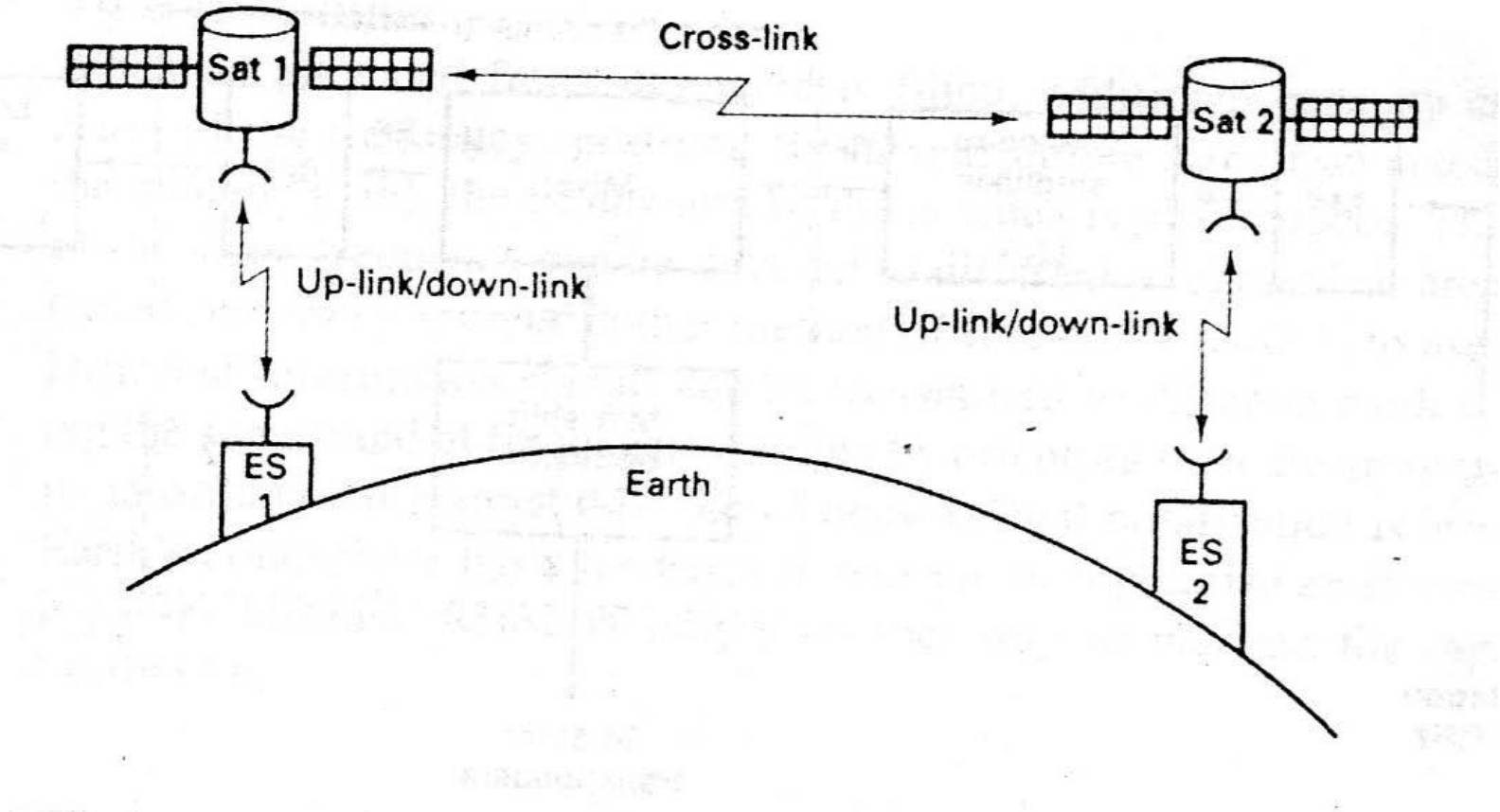
Satellite System Link Models

- Downlink model



Satellite System Link Models

- Cross Link



Satellite System Link Models

- Antenna Look Angles
  - Section 14-6
    - Self Study

System parameters

- Transmit Power and Bit Energy
  - Energy per bit $E_b = P_t T_b = \frac{P_t}{f_b}$
  - Noise power $N = KTB$
  - Noise Density $N_0 = \frac{N}{B} = KT$
  - Energy of bit to noise density ratio $\frac{E_b}{N_0} = \frac{P_t / f_b}{N / B} = \frac{P_t}{N} \times \frac{B}{f_b}$
  - Signal to noise ratio or Carrier Signal to noise ratio $SNR = \frac{P_t}{N}$